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BIOMEDICAL LIMITED

INSTALLATION QUALIFICATION (IQ)

REPORT
FOR
WATER FOR INJECTION STORAGE TANK (4000 LITRES)

EQUIPMENT SR. NO. –

EQUIPMENT ID. –

LOCATION – INFUSION WATER TREATMENT PLANT

PROTOCOL No. – BML/PDF/WT/PT/26/001

REPORT No - BML/PDF/WT/RP/26/001

PROVIDED BY:

BIOMEDICAL LIMTED

KWARA STATE, NIGERIA

CONTROL COPY

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1.0. REPORT FORMAT APPROVAL:

Signatures in this page indicate that this document and its format have been prepared, checked and approved by the personnel concerned. It shall be signed and dated as shown below.

	WRITTEN BY:	**REVIEWED BY:**
Signature		
Position		
Date		

	APPROVED BY:	AUTHORISED BY:
Signature		
Position		
Date		

2.0 OBJECTIVE

The purpose of the study is to ensure proper installation of the two **WATER FOR INJECTION STORAGE TANK (4000 LITRES)** at **THE WATER TREATMENT PLANT, INFUSION FACTORY, BIOMEDICAL LIMITED.**

<u>This study verifies:</u>

- Qualification of the equipment, reinstallation and to ensure that all background information is performed to assure conformance to **BIOMEDICAL LIMITED** and manufacturer's specifications.
- The equipment meets the current good manufacturing practices requirements and all other regulatory obligations.
- No unauthorized or unrecorded modifications have taken place.
- Operation procedures/preventive maintenance procedures have been identified and listed.
- All critical instrumentation has been identified.
- All utilities are properly connected.
- All safety features are accounted for.
- All relevant data have been checked

3.0 SCOPE

- This report is applicable in carrying out the installation qualification of the two waters for injection storage tanks (4000 liters) at the Infusion manufacturing site, Biomedical Limited, Kwara State.
- To be performed after the completion of the design qualification
- To be performed at the time of relocation.

4.0 RESPONSIBILITIES AND QUALIFICATION TEAM

The following departments shall be involved in:

DEPARTMENT	RESPONSIBILITY
Engineering Team	Provision of needed utility and machine installation
Head Of Engineering	Review of the Installation Qualification Document
QA Manager	Final Documentation and system acceptability
Superintendent Pharmacist/factory Manager	Authorization

The following personnel will be involved in the execution of the Qualification

S/No.	DEPARTMENT	NAME	SIGNATURE
1			
3			
4			
5			

5.0 EQUIPMENT IDENTIFICATION AND DESCRIPTION

IDENTIFICATION

Equipment Name	WATER FOR INJECTION STORAGE TANK (4000 LITRES)
Manufacturer's Name	SAHANI ENGINEERS PVT LIMITED (INDIA)
Supplier's name	VITRA VENTURES (INDIA)
Location of Installation	WATER TREATMENT (INFUSION FACTORY)

DESCRIPTION:

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6.0 PREREQUISITE

The equipment shall confirm the specifications and requirements as specified in the Design Qualification

S/N	ACCEPTANCE CRITERIA
1	User Requirement specification
2	Design Qualification

7.0 ABBREVIATIONS AND DEFINITIONS

Abbreviation	Definition
SOP	Standard Operating Procedure
MOC	Material of Construction
HMI	Human Machine Interface
ENGR	Engineering
IQ	Installation Qualification
QA	Quality Assurance
VAC	Voltage Alternating Current
S. S	Stainless Steel
LTD	Limited
°C	Degree Celsius
V	Voltage
GMP	Good manufacturing practices
AC	Alternating Current
Amps	Ampere
W	Watt
SMPS	Switched mode power supply
VDC	Voltage Direct Current
DC	Direct Current

PVC	Polyvinyl Chloride
MCB	Miniature Circuit Breaker
MCCB	Molded case circuit breaker
KW	Kilowatt
PLC	Programmable Logic Control

8.0 REFERENCES:

The list of reference documents, which are referred for preparation and review of this document, is as follows: -

- S.O.P for Installation, Operation & Performance Qualification of equipment & Instrument: BML-QDG-QA-111
- Machine Manual supplied by SAHANI ENGINEERS PVT LIMITED
- Design Qualification supplied by SAHANI ENGINEERS PVT LIMITED

9.0 EQUIPMENT INSTALLATION CHECKLIST:

Pre-Installation Consideration

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	There is no observable physical damage to the tank and all the contact parts are free from scratches & damages.			
2	Sufficient room for servicing, handling & maintenance is provided (around the machine)			
3	Check the floor space. It should not be inclined.			

Assembly of non-assembled items

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Assemble/connect the tank vent filters, gauges, light bulb, Sensor and ultrasonic sensor			

Utilities

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Power Requirement (Three-phase, 5 wires, L1, L2, L3, Neutral & Earth 415VAC ±10% 50Hz)			
2	Adequate illumination at the Proposed Machine room			
3	Proper Earthing System available			

Equipment compliance

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Tank name plate present			
2	Tank Capacity (4000 liters)			
3	Tank MOC: SS316L Contact part, SS304 Non-Contact Part			
4	Tank Serial Number Present (SE86/SE87)			

Safety compliance

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	All external surfaces are free of sharp edges.			
2	Safety earthing arrangements are available.			
3	Emergency Stop Buttons available			
4	Phase indicator is available			

Electrical diagram check

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Verify the electrical panel wiring diagram			
2	Check whether the position of each component in the electrical circuitry is correct			
3	Check whether the wiring of the electrical panel is consistent with electrical diagrams			
4	Check if the electrical diagram is attached to the operation manual			

List of Main Components with their material of construction

S/N	Component	Origin/ Configuration	Qualified / Not Qualified	Checked By	Verified By
MECHANICAL PART LIST					
1	Tank	SAHANI ENGINEER PVT LTD (Main SHELL DISH: SS316L, JACKET SHELL DISH: SS304, LEGS: SS,304, OUTLET VALVE: SS316L, MAN HOLE: SS316L LIGHT: Screw bulb 220VAC, ELECTRICAL PANEL: SS304			
2	Sanitary valve	SS316L Flush bottom Valve 2’’			

3	Air Vent	MATERIAL: S.S316L, TRI CLAMP MOC: S.S304, FILTER: 0.22-micron, Diameter 2'' LENGTH 5'',			
4	PUMP	Motor Manufacturer: Hindustram, 415V, 3.7KW/5 H.P, 7.2Amps, 2900RPM, 0.84PF, 50Hz			
5	Pressure Gauge	SHREEJI (MOC: S.S316L, THREADING SIZE: ¼'' BSP, (0-100Psi/0-7Kg/cm²)			
6	Emergency Button	Manufacturer: TEKNIC, Details: S2 NC Contact, Ratings: A600, IEC947-5-1, EN60947-5-1, 15-240V, 3A			
7	Thermocouple	RADIX (Type: PT 100, Range: -50°C to 250°C, sheath Length: 150mm, sheath Diameter: 6mm, Connection: ½'' BSP)			
8	Pressure relief Valve	Angle type SS304, Connection Type: ¼''BSP			
9	SMPS	Manufacturer: Shavison, Model: G41-60-24, INPUT: 230VAC,1.0A, 50/60Hz, OUTPUT: 24VDC, 2.5A			
10	Miniature Circuit Breaker	(1) Manufacturer: Siemens, Model: 3 POLES, 5SL63167RC, RATINGS: 16Amps (2) Manufacturer: Siemens, Model: 2 POLES, 5SL62107RC, RATINGS: 10Amps			
11	Control Panel Cooling Fan	Manufacturer: Rexnord, MODEL: EC2-12038 S2 W, RATING: 230VAC, 50/60Hz, 0.05Amps, 4.5W			
12	PLC	(1) Manufacturer: DELTA, Model: DVP-12SA211R, POWER UNIT: 24VDC, 1.8W, OUTPUT MODULE: 1.5A, 250VAC 50/60Hz Resistive load. 12SA211RW24280812, V3.02A3 (2) Manufacturer: DELTA DVP-04AD-S2, POWERINPUT:24VDC,			

		2W, I/O RANGE: -10V--P+10V OR -20mA--+20Ma. 04ad-s20w24270996, v4.18a1			
13	Human Machine Interface	Manufacturer: Delta Electronics Inc, MODEL: DOP-107BV, Input: 24VDC, CLASS: 2/340Ma (MAX)			
14	Rotary Switch	Manufacturer: TEKNIC, Details: S2 NC Contact, Ratings: A600, IEC947-5-1, EN60947-5-1, 15-240V, 3A			
15	Hooter	Manufacturer: TNI, Model Number: LB22-D24 Ratings: 24VDC, Mounting Diameter 22mm, Compliance: IEC60947-5-1			
16	Ultrasonic Level Transmitter	Model: ULT-PP-1/ -FP Range: 0- 04Mtr Housing: ABS Plastic Display: LCD Display Power Supply: 24 VDC Output: 420mA (2-wire) Cable Entry:PG11 Diaphragm: PVC Process Connection: G2 (BSP) Protection: IP-66 Process Temp.: -20°C to +80°C Accuracy: ±0.5% Beam Angle: 12.5° Serial No.: 25010028, 25010030			
17	Contactor	Manufacturer: SIEMES, Model: 3TF31, 240VAC Coil			
18	Thermal overload Relay	Manufacturer: SIEMENS, Rating: 4—6.3Amps			
19	DIN Rail Mounted Fuse Holder	Manufacturer: CONNECTWELL, MODEL: CF4U/CF4UL, 0.2—4mm, 1000v, 6.3Amps			
20	Relay Module, with Relays	Relay module Manufacturer: Agility Relay Manufacturer: Omron, ratings: G5Q-14, Coil voltage: 24VDC, contact rating: 10A at 250VAC or 5A at 30vdc			
21	Temperature Transmitter	Manufacturer: Shavison, input: PT-100, Range: 0-400°C, Output: 0-10VDC, Isolation: 3 Points, Order Code: AS746-311, Auxiliary Supply: 24VDC			
22	Light Indicator	Manufacturer: TNI, Model Number: LP22-A230 Ratings: 230VAC, Mounting			

		Diameter 22mm, Compliance: IEC60947-5-1			
23	Main Switch (ON/OFF)	Manufacturer: Salzer, Ratings: 16A-S16-61197, 3-POLES			

10.0 INSTALLATION QUALIFICATION REPORT:

Attachment of all reports written as:

#Attachment 1

#Attachement 2 etc.

Date of Installation:

Date of Completion:

11.0 DEVIATION REPORT

Prepare a Deviation report including the justification of acceptance and impact on the function of the system/equipment

DEVAITIONS:
JUSTIFICATION FOR ACCEPTANCE:
IMPACT ON OPERATION:

Written By: **Date:**

Reviewed By: **Date:**

Approved By: **Date:**

12.0 CONCLUSION

13.0 List of Attachment

Attach all completed, signed data record forms

14.0 REPORT APPROVAL

Written By: **Date:**

Reviewed By: **Date:**

Approved By: **Date:**